

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460164.8444 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.049
Transit depth (Rp/Rs)^2	2.74 +/- 1.63 [%]
Orbital Inclination (inc)	89.97 +/- 1.51
Airmass coefficient 1 (a1)	1.065 +/- 0.017
Airmass coefficient 2 (a2)	-0.034 +/- 0.072
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	#2 - [177, 108]
Optimal Aperture	4.03
Optimal Annulus	5.47
Transit Duration (day)	0.0839 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.049 vs 0.14762 (deviation 0.34σ)
Duration fit vs published	0.0839 d vs 0.1033 d (deviation 3.76σ)
Mid-time O–C	4.0 min
Depth SNR	3.1
Residual scatter	1.6 %

Light curve

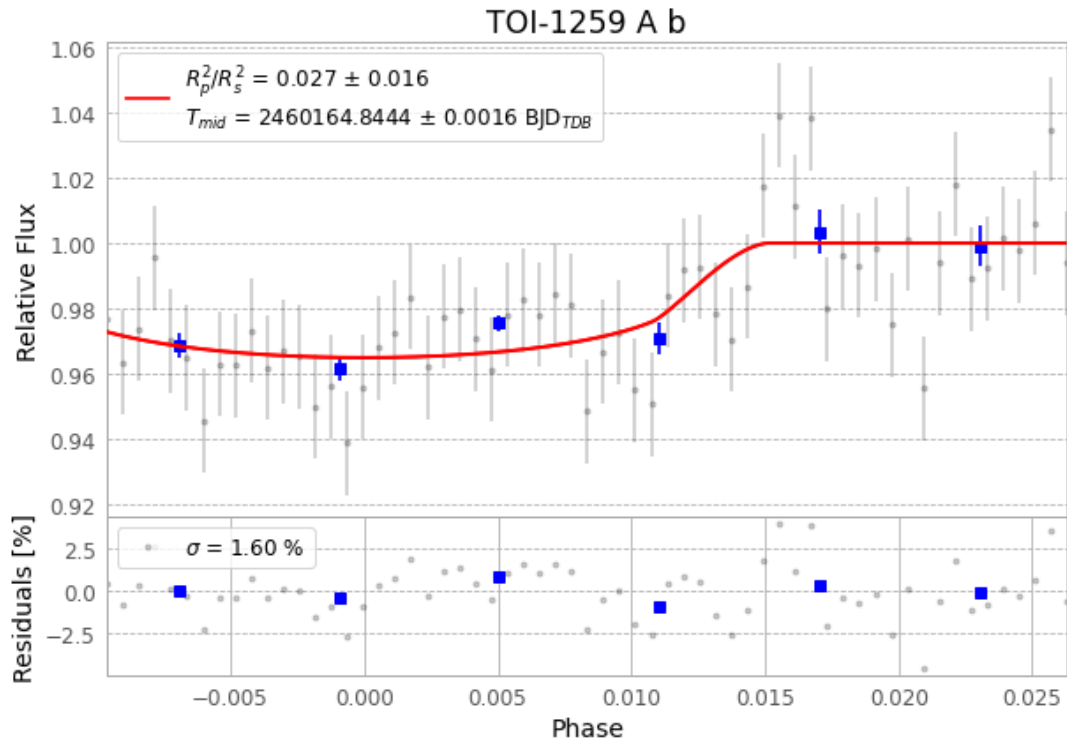


Figure 1 — FinalLightCurve_TOI-1259 A b_2023-08-08.png (SHA-256 5f17739265813f33...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [301, 175], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2b2cfeaf08c7ab6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

61 FITS frames (40 MB), TOI1259230808072715.FITS → TOI1259230808102715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1259-230808.log (f1549f03885a...), FinalLightCurve_TOI-1259 A b_2023-08-08.png (5f1773926581...), FinalLightCurve_TOI-1259 A b_2023-08-08.csv (ea117732afda...)

Compute resources

Wall time 141.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:21Z.